ADVANCING EARTH AND SPACE SCIENCE

Space Weather

RESEARCH ARTICLE
10.1029/2022SW003330

Special Section:
Starlink Satellite Losses during the February 2022 Geomagnetic Storm Event: Science, Technical and Economic Consequences, Policy, and Mitigation

Key Points:

- Geomagnetic storms can cause large variations in neutral density and satellite drag, especially at Very Low Earth Orbit altitudes
- Data assimilative empirical model results and observations during the Starlink event show 20%–30% neutral density enhancements at 210 km
- Full-physics data assimilative models and real-time measurements of thermospheric conditions can mitigate impacts on low Earth orbit operations

Correspondence to:
T. E. Berger,
thomas.berger@colorado.edu

Citation:
Berger, T. E., Dominique, M., Lucas, G., Pilinski, M., Ray, V., Sewell, R., et al. (2023). The thermosphere is a drag: The 2022 Starlink incident and the threat of geomagnetic storms to low earth orbit space operations. *Space Weather*, *21*, e2022SW003330. https://doi.org/10.1029/2022SW003330

Received 22 OCT 2022
Accepted 9 MAR 2023

Author Contributions:
Conceptualization: T. E. Berger, M. Pilinski, J. P. Thayer
Data curation: M. Dominique, E. K. Sutton
Formal analysis: M. Dominique, G. Lucas, R. Sewell, E. K. Sutton, E. Thiemann
Funding acquisition: T. E. Berger, J. P. Thayer

The Thermosphere Is a Drag: The 2022 Starlink Incident and the Threat of Geomagnetic Storms to Low Earth Orbit Space Operations

T. E. Berger[1], M. Dominique[2], G. Lucas[1,3], M. Pilinski[1,3], V. Ray[1,4,5], R. Sewell[3], E. K. Sutton[1], J. P. Thayer[1], and E. Thiemann[3]

[1]Space Weather Technology Research and Education Center (SWx TREC), University of Colorado at Boulder, Boulder, CO, USA, [2]Solar-Terrestrial Centre of Excellence, Royal Observatory of Belgium, Uccle, Belgium, [3]Laboratory for Atmospheric and Space Physics, University of Colorado at Boulder, Boulder, CO, USA, [4]Colorado Center for Astrodynamics Research, University of Colorado at Boulder, Boulder, CO, USA, [5]Now at Kayhan Space, Boulder, CO, USA

Abstract On 03 February 2022, SpaceX launched 49 Starlink satellites, 38 of which re-entered the atmosphere on or about 07 February 2022 due to unexpectedly high atmospheric drag. We use empirical model (NRLMSIS, JB08, and HASDM) outputs as well as solar extreme ultraviolet occultation and high-fidelity accelerometer data to show that thermospheric density was at least 20%–30% higher at 210 km relative to the 9 days prior to the launch due to consecutive geomagnetic storms related to solar eruptions from NOAA AR12936 on 29 January 2022. We model the orbital altitude and in-track position of a Starlink-like satellite in a low-drag configuration at 200 km during minor (G1) and extreme (G5) geomagnetic storms to show that an extreme storm would have at least a factor of two higher impact, with cumulative in-track errors on the order of 10,000 km after a 5-day duration extreme storm. Comparison of the JB08 and NRL MSIS models relative to the HASDM model during modeled historical minor and extreme geomagnetic storms shows that in-track errors on the order of 100 km per day at 250 km, decreasing to cumulative errors on the order of 1 km per day at 550 km during geomagnetic storms. We conclude that full-physics, data assimilative, coupled models of the magnetosphere and upper atmosphere, as well as new operational satellite missions providing “nowcasting” data to launch controllers, space traffic coordinators, and satellite operators, are needed to prevent similar—or worse—orbital system impacts during future geomagnetic storms.

Plain Language Summary On 03 February 2022, SpaceX launched 49 Starlink satellites into staging orbits at 210 km above sea level prior to raising them to their operational altitudes of 550 km. The pre-launch space weather briefing included no information about an ongoing geomagnetic storm. Excessive atmospheric drag due to the geomagnetic storm resulted in 38 of the satellites re-entering the atmosphere on or about 07 February 2022. We use both models and direct measurements of the atmospheric density during the event to show that density values were enhanced by 20%–30% at the 210 km staging altitude relative to values prior to the geomagnetic storm onset, while they were enhanced 90%–160% at higher altitudes. We recommend improving our ability to model the upper atmospheric response to geomagnetic storms to provide accurate forecasts and actionable “nowcasts” of conditions in low Earth orbit to launch controllers, space traffic managers, and satellite operators.

1. Introduction

At 18:13 UT on 03 February 2022, SpaceX launched 49 Starlink broadband communications satellites into a 210 × 350 km staging orbit for on-orbit assessments prior to raising the satellites to their operational circular 550 km orbits. The launch took place during the recovery phase of a minor geomagnetic storm rated G1 on the NOAA Space Weather Prediction Center G-scale. According to SpaceX, “atmospheric drag [increases] up to 50% higher than during previous launches” was experienced in the staging orbit. The increased drag on the Starlink vehicles led operators to command them into a low-drag configuration in an attempt to decrease altitude loss rates. Despite this mitigation, 38 of the 49 satellites could not be recovered from the low-drag configuration for orbit raising and subsequently re-entered the atmosphere and burned up on or about 07 February 2022.

The loss of the Starlink satellites was noteworthy for several reasons. Most notably, this is the first public record of a simultaneous loss of a large number of operational satellites in low Earth orbit (LEO) due solely to atmospheric

Investigation: E. K. Sutton
Methodology: M. Dominique, G. Lucas, M. Pilinski, E. K. Sutton, J. P. Thayer, E. Thiemann
Project Administration: T. E. Berger, J. P. Thayer
Resources: T. E. Berger
Software: M. Dominique, G. Lucas, R. Sewell, E. K. Sutton, E. Thiemann
Supervision: T. E. Berger, J. P. Thayer, E. Thiemann
Validation: M. Dominique, E. Thiemann
Visualization: E. K. Sutton, E. Thiemann
Writing – original draft: T. E. Berger, E. Thiemann

drag effects. Prior reports of total loss of operational satellites during geomagnetic storms have been limited to severe G4 or extreme G5 storms and attributed to solar energetic particle (SEP) radiation impacts. For example, the loss of the ADEOS satellite during the extreme "Halloween" G5 geomagnetic storm in 2003 was due to multiple avionics system failures following charged particle radiation exposure (Rosen et al., 2004). Second, the impact on orbital operations relative to the severity of the geomagnetic storm was surprising: impacts from G1 geomagnetic storms are typically limited to minor perturbations of orbital altitudes that are easily corrected. However, the very low staging altitude of the Starlink satellites, which was intentionally chosen to ensure that malfunctioning satellites re-enter in days rather than in decades, meant that the absolute atmospheric drag increase was significantly larger than expected by the Starlink operators based on experience gained in previous deployments. Third, official pre-launch forecasts of space weather conditions are currently limited and provide no information on ongoing (or incoming) geomagnetic storms. Official Cape Canaveral Space Launch Complex weather and space weather forecasts are issued by Space Force Delta 45, 45th Weather Squadron. Currently, the only space weather item in the forecast is "Solar Activity," included under the "Additional Risk Criteria" section of the forecast.

To the latter point, forecasting thermospheric neutral density perturbations due to geomagnetic storms is complicated by the fact that the storms occur significantly after the root-cause solar magnetic eruption or solar wind high-speed stream generation (Gopalswamy, 2008). An accurate and reliable forecast of thermospheric neutral density during geomagnetic storms therefore requires a linked chain of models from the Sun to the upper atmosphere of the Earth, for example, a full-physics model of the ionosphere-thermosphere-mesosphere (ITM) system coupled on the lower boundary to a tropospheric and stratospheric general circulation model and on the upper boundary to a full-physics model of the magnetosphere and plasmasphere, the regions of the Earth system which process the CME energy and plasma inputs. Development of this complex coupled model system is a goal of applied space weather research (Gombosi et al., 2021), and research versions of these models exist (e.g., Pham et al., 2022; Pulkkinen et al., 2021), but an *operational* fully coupled space weather forecast model for the orbital space environment is still lacking. The term "operational" in this context implies a model that is validated and verified through a full range of space weather conditions, for example, from minor to major geomagnetic storms, is capable of running with noisy near-real-time input data, and rarely, if ever, crashes due to code-based errors (bugs). In particular, none of the coupled research models incorporates data assimilative corrections to the raw model output states and are thus useful only for retrospective analysis of events. The NOAA Space Weather Prediction Center (SWPC) has developed an operational ITM model (the Whole Atmosphere Model—Ionosphere Plasmasphere Electrodynamics, WAM-IPE, Akmaev, 2011) but it also lacks data assimilation and produces a 3-day forecast product that remains unvalidated for a full range of space weather conditions. Therefore, even if the Starlink launch operators had sought up-to-date LEO environment forecasts, the space weather enterprise is currently incapable of providing accurate forecasts.

However, there are orbital systems that have demonstrated the capability to provide an accurate near real-time (NRT) "nowcasts" of LEO thermospheric density conditions in a research setting. In addition, the US Space Force (USSF) High-Accuracy Satellite Drag Model (HASDM; Storz et al., 2005) provides an accurate nowcasting capability via real-time assimilation of some 75 orbital trajectories (see Section 3.1). Thus, there are current or near-term solutions available to launch operators for more actionable *nowcasts* of conditions in LEO.

Here we examine the space weather events that led up to the Starlink losses on 03 February 2022, emphasizing the complexity of the event chain from the Sun to the ITM system, and demonstrating a research measurement of thermospheric neutral density that may serve as a prototype for the NRT nowcasting capability mentioned above. Previous papers have examined the space weather and thermospheric conditions leading up to and including the loss event using both models (Dang et al., 2022 using the TIE-GCM model; Kataoka et al., 2022 using the GAIA model; Fang et al., 2022 using the WAM-IPE model; and Lin et al., 2022 using the MAGE model) and observational data analysis (Lin et al., 2022 using SWARM satellite data; and Y. Zhang et al., 2022 using DMSP/SSUSI data). This paper contributes both new model and observational data analysis of the event which find different values for the thermospheric neutral mass density than some of the earlier studies noted above. In addition, we model thermospheric density conditions during both minor and extreme geomagnetic storms using historical data and a representative model of the Starlink satellites in their low-drag configuration to show that during an extreme G5 geomagnetic storm, the orbital impacts would have been far more severe at all LEO altitudes, not just the low LEO staging perigee altitude of 210 km. We conclude the paper with a commentary and a call to develop the needed space weather forecasting and nowcasting systems, echoing the commentary in Hapgood et al. (2022), so that launch and orbital operations can be protected from the severe and extreme geomagnetic storms that are likely to occur as we approach the solar maximum period of Solar Cycle 25 in the coming years.

2. The 29 January 2022 Solar Magnetic Eruption and Subsequent Interplanetary Conditions

The geomagnetic storm that impacted the SpaceX Starlink launch originated as a large solar magnetic eruption from NOAA Active Region (AR) 12936 (McIntosh classification Dkc; Hale classification β; heliographic position N17 E11) on 29 January 2022. The eruption caused an M1.1 magnitude X-ray flare that peaked at 23:29 UT and a halo CME series (CACTus, Robbrecht et al., 2009, CMEs 2022–0084, 0085, and 0086) detected in SOHO/LASCO C2 coronagraph images from 23:36 to 23:48 UT with a peak velocity of 637 km/s. In addition, AR 12396 produced another large C5 flare at 01:29 UT, with an associated halo CME that was first detected at 01:36 UT at a maximum velocity of 791 km/s (CACTus CME 2022–0087). Although CME velocities are difficult to accurately measure from a single coronagraph viewpoint, it is possible that this second, faster, CME merged with the series of earlier CMEs in interplanetary space, perhaps resulting in a significantly strengthened geomagnetic storm response.

Figure 1 shows that a shock wave was detected by the ACE satellite at the L1 Lagrangian point on 01 February at 21:35 UT (Figure 1a). Simultaneous with the sudden density increase, proton temperature increased to over 200,000 K and solar wind speed increased from 350 to 460 km/s. The shock arrival time implies a CME travel time of approximately 70 hr from the peak of the X-ray flare associated with the eruption on 29 January 2022. The inferred mean velocity in the direction of Earth is approximately 600 km/s. The z-component of the interplanetary magnetic field (IMF) in Geocentric Solar Ecliptic (GSE) coordinates ("Bz") measured by the ACE/MAG instrument began systematically decreasing at 16:00 UT on 02 February, reaching a minimum of approximately −18 nT at 08:55 UT on 03 February before abruptly but briefly rotating to +15 nT at 11:10 UT (Figure 1b). The y-component of the IMF, By, turned negative around 07:00 UT and reached a maximum negative value of −16 nT around 10:45 UT on 03 February, staying largely negative through 05 February, except for brief periods late on 03 February and around 16:00 UT on 04 February. The IMF By direction influences the dayside ionospheric convection patterns that can affect thermospheric expansion in the dusk sector in which the Starlink satellites were launched (e.g., Crowley et al., 2010; Liu et al., 2020; Lu, 2017).

The geomagnetic storm triggered by the CME arrival at Earth commenced at 00:00 UT on 03 February, as indicated by the sudden change in Bz measured at the GOES 16 and 17 satellites in geosynchronous orbit (Figure 1c) and initiation of a decrease in the Dst value (Figure 1d). The planetary "ap" value reached a maximum value of 56 at 09:00 UT, with Dst reaching a minimum value of −66 nT at 10:00 UT on 03 February. The maximum Kp value for the storm was 5+ (NOAA G1 storm), occurring during both the 09 and 12 UT periods. The recovery phase of the storm lasted until approximately 00:00 UT on 04 February, implying a total storm duration of roughly 24 hr. The Starlink launch at 18:13 UT on 03 February occurred on the dusk side of the ionospheric convection pattern during the recovery phase of the storm, with a Dst value of approximately −26 nT and ap value of 15. Note that prior to this main storm, there was a very minor geomagnetic storm on 31 January 2022 with a minimum Dst of −44 nT (max ap = 15) and Kp values of 4 were recorded during four periods prior to the storm onset on 03 February. This shows that the thermosphere was likely pre-conditioned to a stronger response to the CME arrival on 01 February.

Immediately following the launch-day geomagnetic storm, a second G1 geomagnetic storm commenced, again reaching a peak ap value of 56 at 15:00 UT on 04 February and a minimum Dst value of −61 nT at 20:00 UT. The Kp value reached 5+ (NOAA G1) during the 18 and 21 UT periods of 04 February. This storm is somewhat surprising in its intensity since its driver event does not clearly stand out in the solar wind data. It may be due to the arrival of a minor CME system at 22:50 UT on 03 February as indicated by a sudden rise in density and temperature accompanied by a slight rise in solar wind speed from approximately 520 to 580 km/s. The IMF Bz rotated smoothly to negative values over approximately 8 hr from 03 to 11 UT on 04 February, one of the indications of an interplanetary magnetic flux rope (e.g., Palmerio et al., 2018), reaching a minimum of −10 nT at 17:45 UT. This storm eventually recovered to zero Dst around 20:00 UT on 05 February. This storm is relevant to the Starlink loss event since it occurred during the period when controllers were struggling to recover from the impacts of the first storm.

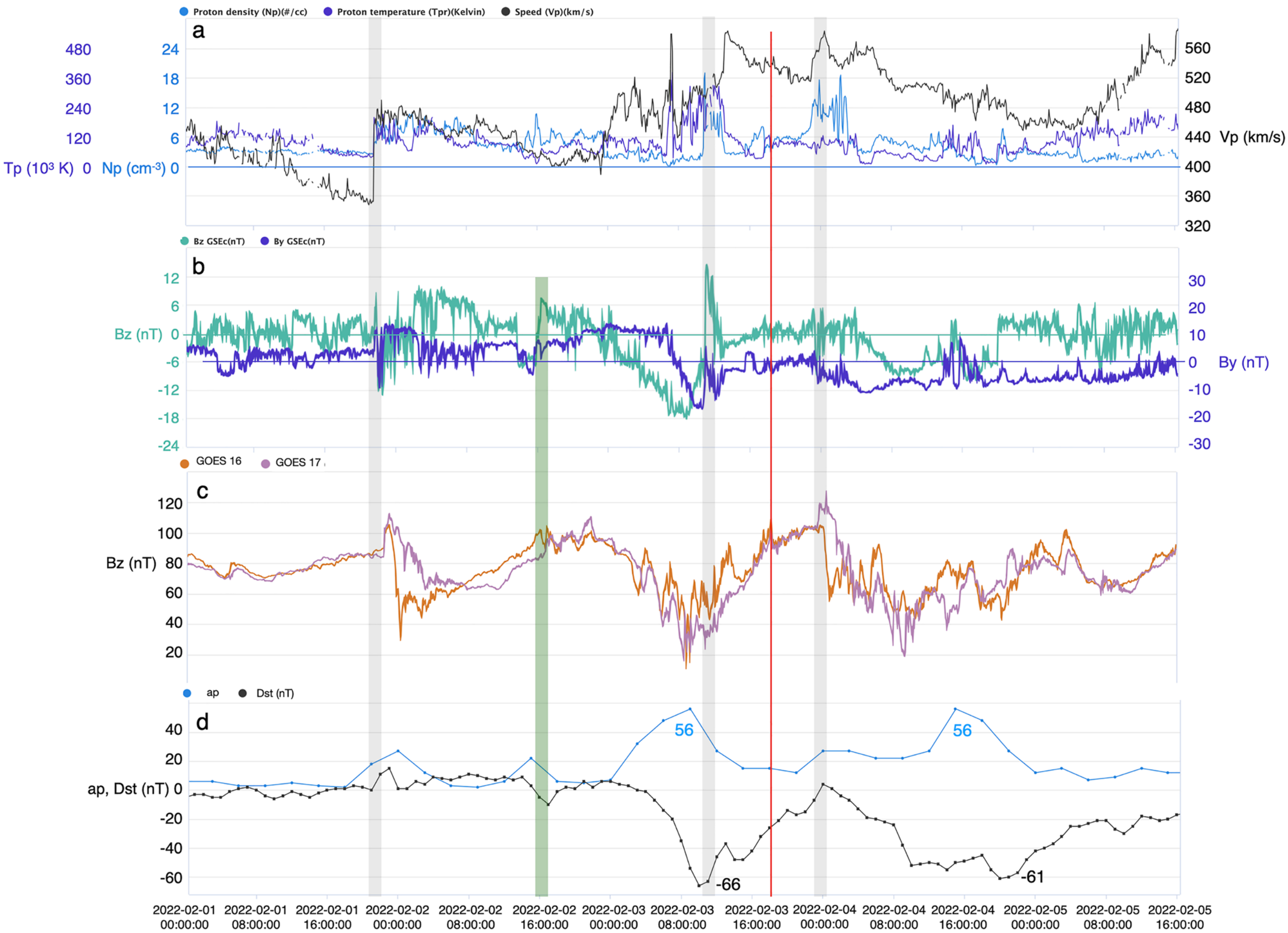

Figure 1. (a) Solar wind density (blue), temperature (purple), and speed (black) measured by the ACE/SWEPAM instrument in an L1 Lagrangian halo orbit from 01 February 2022 to 05 February 2022. Likely CME arrivals indicated by sudden increases in solar wind density and/or speed are highlighted by gray bars. The sudden density and velocity increases on 01 February 2022 at 21:35 UT indicate the arrival of the CME system associated with the 29 January 2022 M1.1 solar flare. (b) ACE/MAG interplanetary magnetic field components By (purple) and Bz (green) in units of nT in GSE coordinates. Bz begins to decrease around 16:00 UT on 02 February 2002 (green vertical bar), reaching a minimum of approximately −18 nT at 08:45 UT on 03 February 2022. Note the zero-line offset between By and Bz. (c) GOES 16 (orange) and GOES 17 (purple) 1-min averaged magnetometer data showing the Bz component of the geo-equatorial magnetic field in nT in GSE coordinates. Two of the apparent CME arrivals (gray bars) coincide with the GOES satellites orbiting near midnight in the magnetotail region. (d) Planetary ap index (blue) and Dst index (black) and the maximum and minimum, respectively, values (insets) achieved during the period. The red line indicates the approximate time of the SpaceX Starlink launch at 18:13 UT on 03 February 2022. All plots were produced with the SWx TREC Space Weather Data Portal: https://doi.org/10.25980/NMFX-XX89.

It should be emphasized that the geomagnetic storms that occurred on 03 and 04 February 2022 were both minor in all measures of intensity. The unexpected impacts of these minor storms on the Starlink launch is likely due to the fact that prior launches occurred when the Sun was exhibiting low sunspot activity and when there were no significant geomagnetic storms. Figure 2 illustrates this point in plots of the maximum solar, geomagnetic, and thermospheric conditions experienced during all Starlink launches since June 2019. Figure 2a shows orbit averaged thermospheric density at 210 km for a representative equatorial circular orbit as derived from the NRL MSIS2.0 model (Emmert et al., 2020; Lucas, 2021). We use the MSIS2.0 model here due to its simplicity of use and analysis compared to the more involved process of obtaining and analyzing HASDM data which was done only for a short period around the launch event (see Figure 3). Orbit averaged thermospheric density was about 55% higher for the 03 February 2022 launch compared to the average density modeled over all launches. Figure 2b shows the F10.7 cm radio proxy for solar Extreme Ultraviolet (EUV) irradiance which rose significantly over the prior four launches. The F10.7 cm flux during the 03 February 2022 launch was 123 solar flux units (sfu; 1 sfu = 10^{-22} W m^{-2} Hz^{-1}), rising to 126 sfu on 04 February 2022, again about 55% higher than the average over all launches. Solar EUV radiation is the primary source of secular heating of the thermosphere, and the appearance in late 2021 and early 2022 of several large sunspot active regions (including AR 12936)

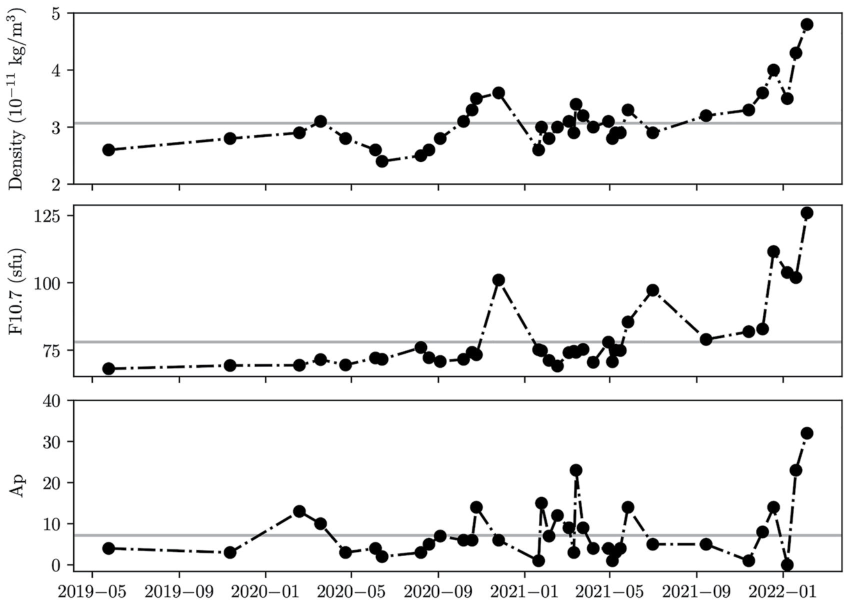

Figure 2. (a) Orbit averaged thermospheric density derived from the NRL MSIS2.0 model; (b) Solar F10.7 cm radio flux in Solar Flux Units (sfu; 1 sfu = 10^{-22} W m^{-2} Hz^{-1}); and (c) the planetary geomagnetic Ap index for the prior 35 Starlink launches leading up to the loss event on 03–04 February 2022. The gray line in each panel denotes the average value over the entire series. The density data show that relative to the earlier Starlink launches in 2022, thermospheric density during the 03 February 2022 launch was approximately 50% higher than the average for previous launches.

led to significant increases in thermospheric heating and concomitant gradual expansion over this period. Thus, when the geomagnetic storms of 03 and 04 February occurred, the thermospheric density was already higher than in prior launch periods due to several months of increased EUV irradiance. Figure 2c shows that except for one launch in March of 2021, no Ap values above 20 had been encountered in prior launches. The launch on 03 February 2022 with a peak Ap value of 32 experienced on 04 February was the first one to take place during significant geomagnetic activity.

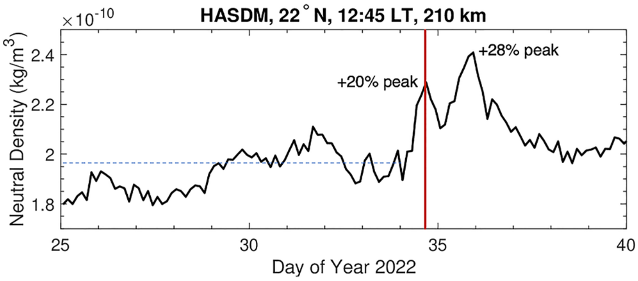

Figure 3. Thermospheric neutral density values inferred from the HASDM data assimilative model at 210 km altitude, 22°N Latitude, and 12:45 Local Time (LT) for DOY 25 to 40, 2022 (25 January–9 February 2022). The latitude and LT values were taken from Two Line Element (TLE) data for one of the Starlink satellites tracked before re-entry. The launch and anomaly occurred on the evening of 03 February (DOY 34, vertical red line), just as inferred density values peaked due to the first G1 geomagnetic storm. Approximate average density value over the 9 days prior to the launch against which the peak density values are compared is indicated by the blue dashed line. The main re-entry event occurred on DOY 38 (07 February). Error bars are not supplied with HASDM model runs but reanalysis establishes an average 8% error range in density estimates over solar cycle timeframes (Tobiska et al., 2021).

3. Thermospheric Neutral Density Response During the 03–04 February 2022 Geomagnetic Storms

3.1. Modeled Neutral Density Response

In this section we present results from the HASDM model (Storz et al., 2005) for thermospheric neutral density values prior to, during, and after the 03 February 2022 Starlink event. The HASDM model is used by the USSF 18th Space Defense Squadron (18th SDS) for orbital trajectory predictions and conjunction assessments. HASDM is based on the JB08 empirical climatological model of the thermosphere (Bowman et al., 2008) and assimilates radar trajectory data from approximately 75 calibration satellites with relatively constant ballistic coefficients at various altitudes to adjust the model outputs to fit the measured trajectory data. Due to this data assimilative capability, HASDM is a significantly more accurate nowcasting model than other empirical or non-data-assimilative full-physics models of the ITM system (Licata et al., 2020; Pilinski et al., 2016). However due to the large-scale averaging and simplification of physics inherent in JB08 (and all other empirical models), HASDM generally cannot provide accurate forecasts during the highly variable conditions associated with geomagnetic storms.

Figure 3 shows thermospheric neutral density values inferred from HASDM from Day of Year (DOY) 25 (25 January) to DOY 45 (14 February) 2022 at

22°N latitude, 12:45 Local Time (LT), and 210 km AMSL: the approximate location of one of the Starlink satellites on DOY 36, as inferred from the orbital Two-Line Element (TLE) data available from the USSF Space-Track orbital catalog (https://www.space-track.org). The figure shows that the launch at 18:13 UT on 03 February (DOY 34) occurred almost exactly at the peak density increase due to the first G1 geomagnetic storm. The peak density at 210 km at this time reached approximately 2.25×10^{-10} kg/m^3, 20% above the average value of about 1.87×10^{-10} km/m^3 for the 9 days prior to the launch. Peak density due to the second geomagnetic storm period occurs near midnight on DOY 35 (04 February) at a value of approximately 2.4×10^{-10} kg/m^3, 28% above the 9-day average prior to the first storm. Notably, the peak of the first geomagnetic storm, as measured by the Ap and Dst indices, occurs around 08:00 UT on DOY 34 (see Figure 1), about *10* hr *prior* to the launch and first density peak. The Starlink launch was during the "recovery phase" of the first geomagnetic storm but coincided exactly with the maximum thermospheric density enhancement from the storm.

The second density peak occurs about 8 hr after the peak Ap and minimum Dst values achieved in the second G1 geomagnetic storm. Although the solar wind conditions that triggered the second storm were significantly weaker than the first, with a shorter and lower intensity negative Bz period, the higher density peak and slightly faster response time associated with this second peak may be attributable to the fact that the thermosphere was significantly disturbed due to the first storm. The decay time of the second peak, as defined by the time from the peak to the valley in modeled density, was about 3 days. The 10- and 8-hr delays between geomagnetic storm peak and thermospheric density increase peak are somewhat longer than the average 6.5 ± 0.75-hr delay found in Berger and Barlier (1981) but are comparable to the 8-hr delay cited in Forbes et al. (1978).

3.2. Measured Neutral Density Response

While the HASDM model is well-calibrated and its error ranges are well characterized (Tobiska et al., 2021), it is of interest to compare HASDM inferences of density increase during the geomagnetic storm to direct measurements at similar altitudes. The Large Yield RAdiometer (LYRA; Dominique et al., 2013) instrument aboard the European Space Agency (ESA) Project for Onboard Autonomy 2 (PROBA2) microsatellite made direct density measurements from ~150 to 330 km during the 03 February Starlink event. LYRA is a radiometer that observes the Sun at high cadence (50 Hz, resampled to 1 Hz for the current study) in four broadband channels in the Soft X-ray (SXR)toUV spectral range. Because its main objective is monitoring and analysis of solar flares, its orbit was chosen to be a Sun-synchronous dawn-dusk polar orbit with an altitude of ~700 km to minimize the number and duration of eclipses. Still, because of a combined effect of the Earth's obliquity and the satellite orbit inclinations, eclipses occur for 3 months every year, from early November to early February. During this period of the year, occultations, that is, the progressive attenuation of the solar signal while the optical path of LYRA crosses deeper and deeper layers of the Earth's atmosphere, special observation campaigns are set up to measure solar occultations using a backup unit of the instrument, which is less subject to aging than the nominal unit. Such a campaign was taking place in the two first months of 2022, except during the period of 31 January 05:20–2 February 18:35, when another special observing campaign was implemented. However, density measurements are available during the week preceding the event, as well as during the entire first G1 geomagnetic storm.

Solar occultations infer atmospheric number density from the attenuation profile of the solar signal as it passes through the atmosphere (Thiemann & Dominique, 2021; Thiemann et al., 2017). The regions scanned during the occultation process are constrained by the orbit of the spacecraft and the DOY. During early February 2022, observation latitudes at dawn varied from 58° at an altitude of 200 km to 51° at an altitude of 300 km, while at dusk the observation latitudes varied from 49° to 39° for altitudes between 200 and 300 km. Figure 4 shows the altitude profiles of thermospheric number density at dawn (a) and dusk (b) measured by LYRA between 26 January and 10 February. Data coinciding with the 03–04 February geomagnetic storm are shown in pink. Note that these plots are calibrated in absolute number density, not mass as in Figure 3, so the values are not directly comparable to the HASDM inferences. The LYRA instrument is optimized for solar irradiance studies and thus does not include atmospheric composition discrimination required to derive mass density. Future EUV occultation instruments will include composition discrimination and hence will provide direct mass density estimates.

Time series at 200 and 300 km of the occultation data are shown in Figure 5. Absolute number density values at the two altitudes are plotted in the top panels, with the DST index overplotted in pink; and the bottom panels show the same data normalized to the daily average of the first day shown. The density increase caused by the first geomagnetic storm is clearly visible starting on 03 February 2022 at both altitudes. Density enhancements over the course of the next 2 days are well correlated with the double-peaked profile of Dst due to the two successive

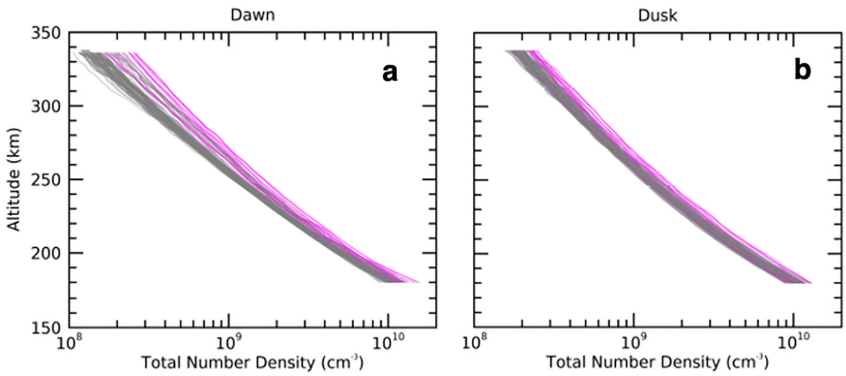

Figure 4. (a) Number density profiles measured by the LYRA instrument on PROBA2 from DOY 26 to 40 (26 January–10 February) 2022 at the dawn (West) terminator; and (b) at the dusk (East) terminator. The pink curves indicate profiles measured from DOY 34–35 (3–4 February), during the geomagnetic storms responsible for the Starlink loss; data from other days is plotted in gray.

geomagnetic storms. The relative density enhancement during the first storm on the dawn side at 200 km altitude and 59°N latitude is approximately 20% (Figure 5c), in good agreement with the HASDM model value for 12:45 Local Time (Figure 3). The second density dawn-side peak on DOY 35 is approximately 33% at 200 km, again in good agreement with the HASDM results that show a 28% density enhancement at that altitude, but again at 12:45 LT. Figure 5d shows the dusk side measurements of relative density increase at 200 km which is at about 10% for the 03 February storm but increases to almost 40% late on 04 February during the second storm, significantly larger than the 28% value estimated by the HASDM model. It is notable that the peak relative density increase measured during the 04 February storm does not show a large altitude variation like the dawn sector measurements in Figure 5c.

The different dawn and dusk density responses have been noted in previous studies (e.g., Thiemann et al., 2017; Y. Zhang et al., 2022) with the inferred magnitude of exospheric temperature sector difference around 100 K (dawn sector hotter). It is worth emphasizing again that density enhancements due to Joule heating from geomagnetic storms increase rapidly with altitude, reaching 90% at 330 km in the dawn sector and 40% at 330 km in the dusk sector in the LYRA measurements. It is also worth emphasizing that these levels of density enhancement were caused by relatively weak G1 geomagnetic storms.

Figure 6 shows the mass density derived from the Gravity Recovery and Climate Experiment Follow-On (GRACE-FO; Kornfeld et al., 2019) accelerometer data taken at altitudes ranging from 482 to 535 km from 2022 DOY 30 to 40 (30 January–09 February). We show these data to illustrate the relative neutral density responses to a G1 geomagnetic storm near the operational altitude of the Starlink constellation (~550 km) and for comparison to derived neutral density responses from SWARM satellite data analysis by Lin et al. (2022). In addition, in the following Section we simulate the relative impact of geomagnetic storm-related neutral density increases at both the Starlink staging altitude and the operational altitude. It is of interest to show relative density data measurements near the operational altitude to compare with both the simulated impact analyses in Section 4 and with other densities inferred at these altitudes during the geomagnetic storms.

GRACE-FO is a pair of low-drag satellites flying in formation in a 482–535 km polar orbit. Comparing Figure 6a and 6b with the mass density values in Figure 3 shows that the density at 500 km is about three orders of magnitude less than the density at 210 km. The density enhancement peak from the first storm at 500 km shows a 94% increase above the average values for the preceding 4 days prior to the launch (dashed blue line) and occurs about 4.8 hr *earlier* than the peak at 210 km which is shown in the HASDM data in Figure 3 to occur very close to the launch time. The density enhancement from the second storm on 04 February reaches 163% relative to pre-launch values at 500 km and occurs roughly at the same time as the peak at 210 km seen in the HASDM data in Figure 3.

4. Comparison of the Impact of a G5 Storm Versus a G1 Storm on LEO Trajectories

The Starlink loss event illustrates that even a relatively minor density increase in VLEO was consequential due to its unforeseen nature. However, in extreme geomagnetic storms such as the 2003 Halloween storms or the

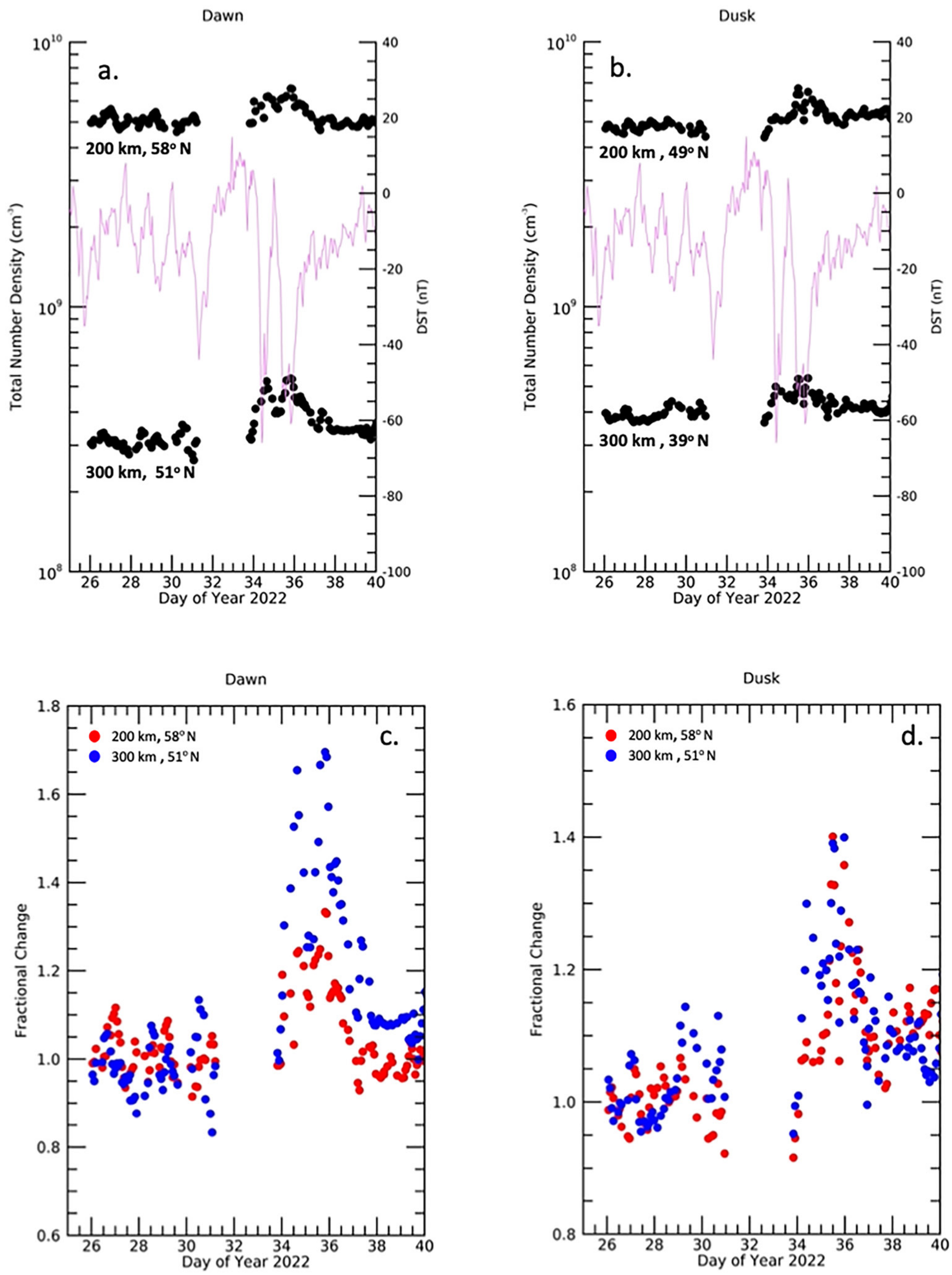

Figure 5. Thermospheric total number density measured by LYRA between DOY 25–40 (25 January–9 February) 2022 at 200, 250, 300, and 330 km is shown in absolute cm^{-3} units (a) and (b) and fractional units (c) and (d). Dawn data are shown in the left panels and dusk data are shown in the right panels. The corresponding latitudes of the observations are given on each panel. The DST index is overplotted (pink curve) in panels (a and b) to indicate the contemporaneous geomagnetic activity.

1989 geomagnetic superstorm, thermospheric neutral density can increase by much larger amounts (e.g., Sutton et al., 2006) and a large portion of the orbital catalog can be invalidated as satellite and debris tracks are "lost" due to large changes in orbital altitude and in-track (i.e., in the direction of travel along the orbit) position (e.g., Burke, 2018). Loss of tracking can lead to downlink and uplink communication disruption as ground antennas fail to acquire satellites that are potentially thousands of km off-track, as well as conjunction estimates that become so uncertain as to be effectively useless, ultimately requiring time-consuming radar re-acquisition of all objects in LEO.

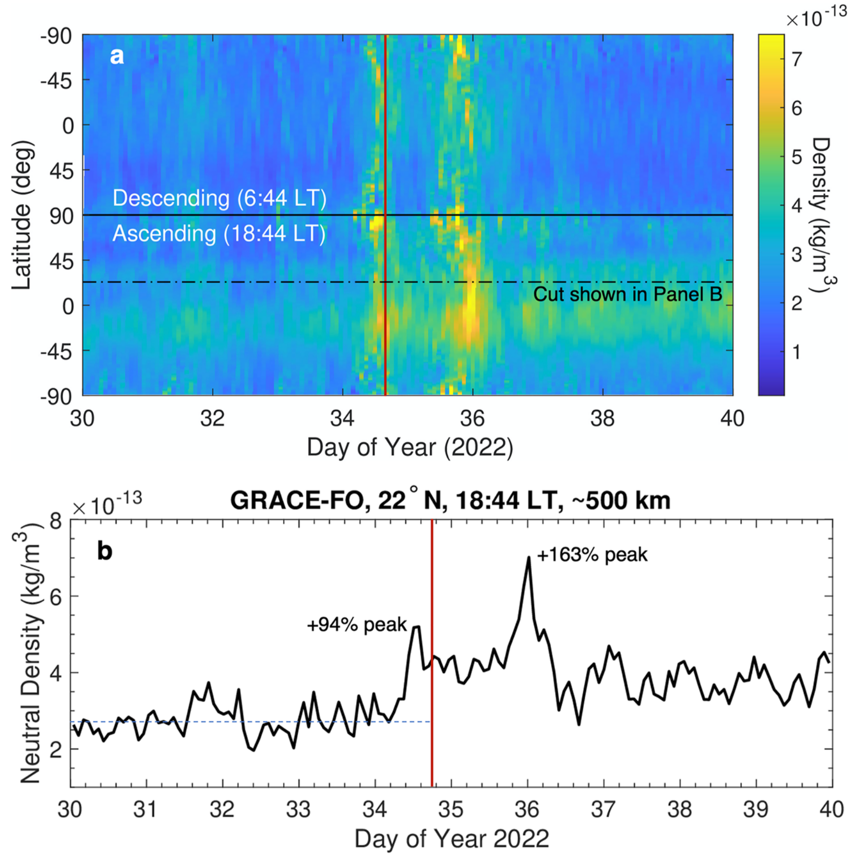

Figure 6. Neutral mass density derived from accelerometer measurements on the GRACE-FO satellites from DOY 30–40 (30 January–9 February) 2022. (a) Neutral mass density as a function of DOY and geographic latitude for the ascending (bottom half) and descending (top half) nodes of the orbit. The color bar indicates mass densities ranging from 1 to 7×10^{-13} kg/m^3 at the average altitude of 500 km over the orbit. The red vertical line shows the approximate time of the Starlink launch at 18:13 UT on DOY 34; the dot-dash line shows the density profile cut plotted in Panel (b). (b) GRACE-FO neutral mass density values along the dot-dash line in Panel a indicating 22°N Latitude and 18:44 UT at an altitude of about 500 km. The density enhancement peak for the first geomagnetic storm occurs approximately 4.8 hr before launch time (red vertical line) at this altitude and reaches 94% relative to the 5 days prior to launch (blue dashed line).

Unfortunately, detailed orbital trajectory information during the 1989 and 2003 storms is not available for analysis. To illustrate the magnitude of the impacts to VLEO and LEO trajectories, we therefore simulate what the impact of an extreme G5 geomagnetic storm would be on a Starlink-like satellite at both the staging altitudes of 200–300 km and the operational altitude of 550 km and compare the magnitude of those impacts to simulated G1 storm impacts.

A model of the Starlink satellite in its low-drag ("open book") configuration is propagated in orbits of various perigee altitudes (250, 350, and 550 km) for two simulated geomagnetic storm conditions: a G5 level storm replicated from the 29 October 2003 storm (the "2003 Halloween storm"), and a G1 level storm replicated from the 02 February 2022 Starlink storm. A high-fidelity force model with 100×100 gravitational field, third-body forces of the Sun and Moon, box-wing drag, and solar radiation pressure is used for orbit propagation (the model is available in the open data repository associated with this paper). The density state for each storm is determined using the HASDM model. The resulting altitude decay is plotted in Figure 7a for a similar orbit as the staging orbit of the Starlink satellites with perigee at 210 km and apogee at 340 km during a G1 and a G5 geomagnetic storms over a 4-day duration. The altitude decay of the Starlink-like satellite at 210 km perigee altitude ranges from 20 km over 4 days for the G1-level storm to around double the amount for the G5 storm over the same duration. These values are relatively small because the "open book" satellite model is a very low-drag configuration; higher frontal area-to-mass ratio configurations will have much larger altitude losses. Also note that we do not include a detailed torque model that can take into tumbling and the much larger ballistic coefficient that this would impart.

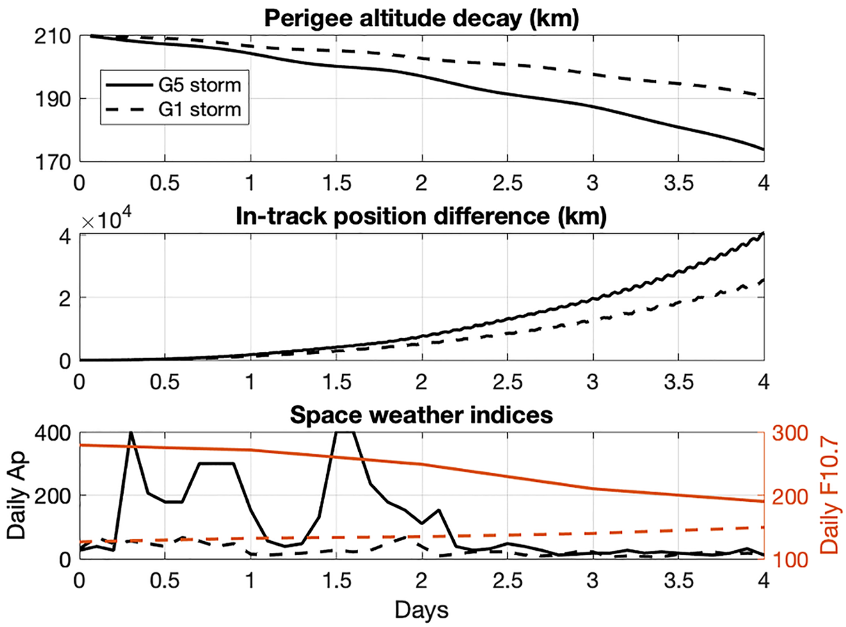

Figure 7. (a) Perigee altitude decay and (b) change in the in-track position as a function of time relative to an orbit unperturbed by atmospheric drag for a "Starlink-type" satellite with a fixed attitude to produce minimum drag force in an eccentric orbit with the perigee altitude at 210 km and apogee altitude at 340 km orbit during a G1 and G5 geomagnetic storm. (c) Planetary geomagnetic Ap index and F10.7 radio flux in solar flux units for the simulated G1 and G5 storms. The solid lines show a representative extreme G5 geomagnetic storm (the "Halloween 2003" storm), and the dashed lines show synthetic data representative of the 03 February 2022 G1 geomagnetic storm.

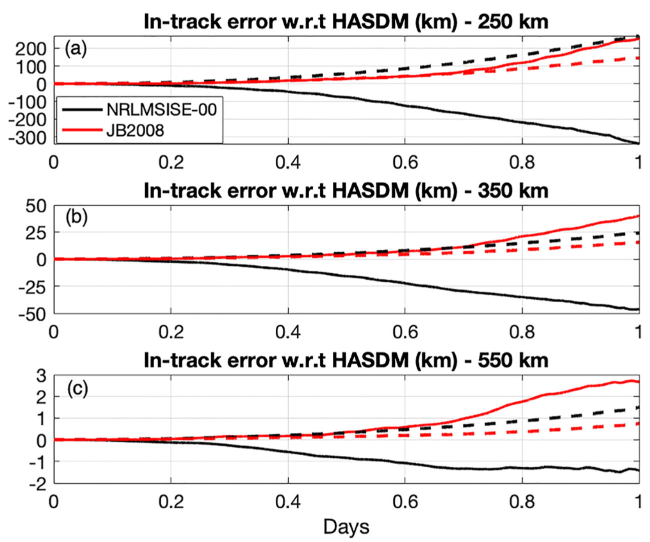

Figure 8. In-track positional error incurred over 1 day when using the MSISE-00 (black) or JB08 (red) models with respect to HASDM as the reference density model during G1 and G5 geomagnetic storms for a "Starlink-type" satellite in low-drag configuration at (a) 250 km; (b) 350 km; and (c) at 550 km. Results for a G5 geomagnetic storm (solid lines) and a G1 geomagnetic storm (dashed lines) are shown. Note that the underlying thermosphere model for HASDM is the JB08 model; the curves for JB08 error relative to HASDM shown above thus reflect the correction of JB08 errors due to data assimilation at any storm level. The errors at 250 km for both models at either storm level are an order of magnitude larger than those at 350 km and two orders of magnitude larger than 550 km.

Figure 7b shows the corresponding changes in the in-track position relative to a reference orbit unperturbed by atmospheric drag. The in-track position change (δs) is calculated along the curved path of the satellite as $\delta s = \delta M \frac{v(t)}{n(t)}$ (Emmert et al., 2017), where δM is the change in Mean anomaly, $v(t)$ is the speed of the satellite, $n(t)$ is the mean motion of the satellite. Figure 7b shows that the cumulative in-track position change is about 25,000 km over the 4-day duration of the G1 storm and increases to approximately 41,000 km, around a 60% change, over duration of the G5 storm. It is important to note that in-track position is advanced relative to the unperturbed orbit since the satellite velocity increases as altitude loss converts gravitational potential energy to kinetic energy—the satellite speeds up.

The orbital changes during geomagnetic storm conditions are important factors for consideration when planning for station-keeping and collision avoidance. This requires predicting the satellite orbits into the future using a given atmospheric density model. Depending on the model used, the orbit prediction can vary significantly. Figure 8 compares the in-track error calculated for the same storms shown in Figure 7, but over only a 1-day period, for the same Starlink open-book configuration using the JB08 and MSISE-00 (Picone et al., 2002) empirical density models at three LEO altitudes. *The values shown are relative to the HASDM model calculations for the same conditions*, that is, the plots show the difference in orbital position when calculated with the JB08 or MSISE-00 model relative to using HASDM. The in-track position difference relative to HASDM at 250 km altitude is between 255 km (JB08) and 340 km (MSISE-00) over the G5 storm duration. At 350 km altitude, the in-track error is between 40 km (JB08) and 50 km (MSISE-00) over the storm duration. The error decreases to between 1.5 and 2 km at 550 km due to the much lower absolute mass densities at that altitude.

Figure 8 confirms that the accuracy of non-data-assimilative empirical density models is significantly diminished during severe geomagnetic storms (Oliveira et al., 2021). Note also that the models can under-predict or over-predict the densities (and hence drag forces and in-track errors) relative to one another, as demonstrated by the difference in signs of the in-track error for MSISE-00 for the two storms. Orbital velocities at 550 km are 7–10 km/s so the in-track position differences at that altitude represent less than 1-s of satellite travel time and may not impact communication links significantly. But the differences at lower altitudes imply that the choice of density model can result in in-track timing errors measured in 10s of seconds which could severely impact antenna pointing calculations. Figure 8 can also be viewed as illustrating uncertainly levels in orbital position introduced by model differences which translate directly into increased uncertainty in conjunction calculations that can significantly impact collision probabilities and resulting maneuver strategies.

Finally, it is important to note that for satellites with on-board GNSS receivers, ionospheric disturbances during extreme geomagnetic storms can lead to loss of lock over significant portions of the orbit (S. Zhang et al., 2020), particularly for polar orbiting satellites at lower LEO altitudes. Thus GNSS-derived satellite states may be unavailable or inaccurate during severe geomagnetic storms and operators may be solely reliant on orbit propagation models of varying accuracies (as shown in Figure 8) to estimate satellite positions for conjunction calculations.

5. Discussion and Conclusions

5.1. Discussion

The peak 20–28% (±8%) relative neutral density increases that we report from HASDM model analysis of the Starlink orbits at the 210 km orbital perigee differ from values reported in other modeling studies of the event. Fang et al. (2022), using the WAM-IPE model, report peak density increases at 210 km relative to a 10-day average prior to the first storm of 50–100%, depending on global location. Similarly, Lin et al. (2022), report 150% peak relative density increases at 210 km using the MAGE model. They also show 50% increases based on NRLMSIS2.0 and DTM-2013 (Bruinsma, 2015) model calculations. Kataoka et al. (2022) use the GAIA model and find local areas near the polar regions of 50% density increases relative to the global average density on 01 February, but increases at all other locations appear to be in the range of 10–20%. In contrast to Lin et al. (2022), they find peak relative density increases using NRLMSIS2.0 of only 25%. Finally, Dang et al. (2022) find a global average density increase of 23% at 210 km relative to the average density on 02 February. We hypothesize that the differences in model results found between studies is due in part to the reference density levels chosen as well as to whether the values reported are globally averaged, averaged over the specific 210 × 350 km staging orbit trajectories of the Starlink satellites, or referring to local structures, for example, in the polar regions.

It should be pointed out that the WAM-IPE and MAGE model results of Fang et al. (2022) and Lin et al. (2022), respectively, are full-physics numerical simulation outputs from post-facto runs that have the advantage of access to data from Starlink (210 × 350 km orbit) and SWARM C (440 km circular orbit) satellites, respectively, for reference. While full-physics models have the potential of significantly outperforming empirical models in forecasting skill, until these models incorporate data assimilative correction, their outputs cannot be directly compared to HASDM model outputs that are data assimilative and therefore provide a more accurate *nowcasting* characterization of conditions in LEO for any given event.

The LYRA and GRACE-FO measurements of relative density enhancement during the Starlink storms that we show in Section 3.2 also differ from values found in other studies. Y. Zhang et al. (2022) analyze DMSP/SSUSI atomic Oxygen 130.4 nm radiance data to infer peak orbit averaged relative density increases at 210 km of 11%–18% (±9%) on the dusk terminator between −30° and ±30° latitude and 40%–59% on the dawn terminator (for the same latitude range) during the 03 February storm. At 520 km, they infer relative density increases over the same latitude range of 18%–26% on the dusk side and 300% on the dawn side for the 03 February storm. Lin et al. (2022) use accelerometer data to infer peak relative density increases along the orbits of the SWARM A and B satellites (440 and 515 km altitudes, respectively) of 87% and 118%, respectively, for the 03 February storm and 105% and 124%, respectively, for the 04 February storm. Finally Dang et al. (2022) infer peak relative density increases using accelerometer data from the SWARM C satellite at 440 km of 50% and 75% for the 03 February and 04 February storms, respectively. Figure 5 shows that our LYRA occultation analysis finds a peak relative density increases at 200 km on the dusk sector of 10% and 40% for the 03 and 04 February storms, respectively. On the dawn sector the increases are 20% and 33% for the same storms. These are higher on the dusk side and

10.1029/2022SW003330

lower on the dawn side than the values inferred at 210 km by Y. Zhang et al. (2022). Similarly, our GRACE-FO analysis at 500 km and 22°N latitude at 18:44 UT (Figure 6b) shows a peak relative density increase of 94% for the 03 February storm. This is much larger than the 18%–26% relative increases found on the dusk side at 520 km by Y. Zhang et al. (2022). On the other hand, it is within the range of the 87%–118% increases found for the 03 February storm in the SWARM A and B orbit averaged data by Lin et al. (2022). The origin of the discrepancy between our results and those of Y. Zhang et al. (2022) may lie in the relatively complex analyses required to infer densities from EUV occultation signals and O 130.4 nm radiances, respectively. However, the level of agreement between the accelerometer measurements and the LYRA inferences demonstrates the potential utility of EUV occultation measurements as near-real-time thermospheric density nowcasting data over the critical VLEO range of about 250–400 km where satellite drag is by far the dominant source of non-conservative forcing.

We emphasize that the range of relative density increases shown above were generated by geomagnetic storms that were minor compared to historical storms of record such as the March 1989 superstorm and the Halloween 2003 extreme events. In these larger storms, the relative density increases were much higher and resulted in orbital impacts to satellites at all altitudes below about 1,200 km. For example, in the 1989 superstorm (Dst < -500 nT), over 2,500 satellites and debris objects in LEO were lost from the Space Surveillance Network (SSN) tracking catalog for several days (Burke, 2018). The lost objects comprised most of the objects in LEO at the time. Similarly, anecdotal accounts of the extreme 2003 Halloween geomagnetic storms indicate that the entire LEO catalog was invalidated for at least 3 days following the chain of geomagnetic storms from 29 to 31 October, requiring emergency operation shifts at SSN radar facilities to reacquire critical space assets. While there were no public reports of collisions due to the loss of the tracking catalog during the 1989 and 2003 events, the LEO orbital environment during those storms was much less congested than it is today.

The orbital simulations illustrated in Section 4 show that had the Starlink launch occurred during a G5 geomagnetic storm, all 49 satellites would likely have been lost due to the much higher drag forces encountered in those conditions. In addition, Starlink satellites at the operational altitude of 550 km, along with most other active satellites and debris in LEO, would have experienced severe orbital trajectory impacts likely leading to the loss of their tracking information in the orbital catalog for several days. Currently there are over 5,500 active satellites and 20,000 debris objects larger than 10 cm tracked in LEO (ESA, 2022). This is significantly more than the approximately 5,000 satellites and debris objects in LEO during the Halloween 2003 storms. If a geomagnetic superstorm were to strike today, or in 5 years when the currently developing LEO megaconstellations are more complete, it is not known if the loss of catalog information for several days to a week or more would cause collisions in the much more crowded LEO environment. Of particular concern is the fact that satellites in megaconstellations are maneuvering autonomously to avoid collisions based on uplinked orbital object catalogs. Are the failure modes of these autonomous systems sufficiently robust to avoid inadvertently causing collisions when the uplinked orbital catalog is missing or, worse, wrong?

5.2. Conclusions

Data assimilative modeling as well as solar occultation and accelerometer measurements during the geomagnetic storms of 03 and 04 February 2022 show that orbit-averaged relative thermospheric neutral density at 200 km altitudes increased by at least 20%–30% over pre-storm levels. These density increases, although not nearly as large as seen in major historical geomagnetic storms, led to control issues due to unexpectedly high drag forces and torques at the 210 km perigee of the staging orbits of the 49 Starlink satellites launched on 03 February 2022.

The Starlink loss event and the ongoing need for more accurate and actionable forecasting and nowcasting information on environmental conditions in LEO mandates better integration of space weather research and operational forecasting as well as improved end-user engagement so that new observations and models can be tailored to their requirements. Our investigations into the Starlink loss event have led to the following conclusions:

1. Near-real-time HASDM Dynamic Calibrated Atmosphere (DCA) output should be integrated into all commercial launch operations to provide what is currently the most accurate thermospheric density nowcasting capability available. *Post facto* HASDM model output is currently available to the research community for use in validating models up to 31 December 2019 (Tobiska et al., 2021). But the real-time HASDM model output should be made continuously available, perhaps with some delay, to the forecasting research community for use in benchmarking the *forecasting* capabilities of the full-physics data-assimilative models that are now in development, for example, the WAM-IPE model at NOAA/SWPC.

2. NOAA should improve the WAM-IPE model by coupling it the operational version of the Space Weather Modeling Framework (SWMF) "Geospace" model (Gombosi et al., 2021) to provide more realistic electric potential and auroral energy inputs from the magnetosphere. This, along with the implementation of a data assimilation method to ensure that the model output states are corrected to reality on a regular cadence, should provide the first accurate, fully validated, and actionable 2–3 day forecast of thermospheric and ionospheric conditions for space weather end-users.
3. Government agencies should work to transition space physics research missions that have demonstrated an ability to provide accurate thermospheric temperature, composition, and density data to a real-time operational capability that supports civil and military launch and LEO orbital operations. In this paper we have demonstrated the capability of EUV occultation measurements from the LYRA instrument on the ESA PROBA-2 satellite to infer accurate thermospheric number densities over a range of critical LEO and VLEO altitudes. An operational EUV occultation satellite mission derived from the LYRA design could add thermospheric composition capabilities, and a constellation of such satellites could provide continuous measurements for both real-time nowcasting information to launch and orbital operators as well as atmospheric profile data for assimilation into ITM forecasting models. Another successful demonstration of technology for global-scale thermospheric monitoring is the NASA Global Observations of the Limb and Disk (GOLD) mission (Eastes et al., 2017). The GOLD team has demonstrated low-latency delivery of thermospheric temperature data to NOAA (Codrescu et al., 2020) and these data have been assimilated into the NCAR WACCM-X ITM research model (Laskar et al., 2021, 2022). By replicating the GOLD instrument on GEO satellites in a variety of LT positions, continuous low-latency delivery of thermospheric temperature and O/N_2 ratio composition data could be provided for nowcasting and assimilation into operational ITM forecasting models. In parallel, developing a research-to-operations (R2O) pipeline of data assimilation capabilities from the WACCM-X research model to the operational WAM-IPE model would accelerate our ability to accurately forecast conditions in LEO.

The new operational measurement solutions discussed above could be developed and deployed in a relatively short period if given sufficient resources. ESA and NASA have demonstrated the technologies and mission scientists have developed the capabilities to rapidly analyze and interpret the data. The challenge lies in transitioning these research mission demonstrations to the operational space weather organizations that are responsible for NRT monitoring and forecasting of the space environment. Such transitions will require close coordination between research and operational agencies as well as dedicated funding for R2O transitions of space weather observational and modeling capabilities.

Data Availability Statement

The data used to produce the figures in the study are available on the University of Colorado at Boulder Space Weather Technology, Research, and Education GitHub page at https://github.com/swxtrec/2022_starlink_event_paper under the permissive MIT license. There are subdirectories for each figure with the corresponding code and data to replicate the figures.

Acknowledgments

LYRA is a project of the Centre Spatial de Liège, the Physikalisch-Meteorologisches Observatorium Davos, and the Royal Observatory of Belgium funded by the Belgian Federal Science Policy Office (BELSPO) and by the Swiss Bundesamt für Bildung und Wissenschaft. HASDM data were accessed via an Orbital Data Request to the 18th Space Defense Squadron of the United States Space Force (USSF) and the resulting analyses shown here are by permission from the USSF. GRACE-FO data were processed as part of the SWARM DISC TOLEOS project and were provided by the European Space Agency. M. D. acknowledge support from the Belgian Federal Science Policy Office (BELSPO) in the framework of the ESA-PRODEX program, Grant 4000134474. J.P.T. and V.R. were supported by the National Aeronautics and Space Administration under Grant 80NSSC21K1554 issued through the Heliophysics Division Space Weather Science Application initiative. J.P.T. and E.K.S. were supported by NASA GDC Interdisciplinary Science Team contract 80GSFC22CA012 to the University of Colorado at Boulder. E.T. was supported by the PROBA2 guest investigator program, ESA-PRODEX PEA No. 4000134474, and by the National Aeronautics and Space Administration under Grant 80NSSC21K0028 issued through the Heliophysics Division Space Weather Science Application initiative. T.E.B. and E.K.S were supported by a Grand Challenge Grant "Our Space. Our Future" by the University of Colorado at Boulder to establish the Space Weather Technology, Research, and Education Center. We thank Jenny Knuth and the SWx TREC Space Weather Data Portal team for facilitating data access and plot creation for Figure 1. Information on data sources used in Figure 1 are available on the Space Weather Data Portal: https://doi.org/10.25980/NMFX-XX89.

References

Akmaev, R. A. (2011). Whole atmosphere modeling: Connecting terrestrial and space weather. *Reviews of Geophysics*, *49*(4), RG4004. https://doi.org/10.1029/2011RG000364

Berger, C., & Barlier, F. (1981). Response of the equatorial thermosphere to magnetic activity analyzed with accelerometer total density data. Asymmetrical structure. *Journal of Atmospheric and Terrestrial Physics*, *43*(2), 121–133. https://doi.org/10.1016/0021-9169(81)90070-2

Bowman, B. R., Tobiska, K. W., Marcos, F. A., Huang, C. Y., Lin, C. S., & Burke, W. J. (2008). A new empirical thermospheric density model JB2008 using new solar and geomagnetic indices. In *AIAA/AAS astrodynamics specialist conference*.

Bruinsma, S. (2015). The DTM-2013 thermosphere model. *Journal of Space Weather and Space Climate*, *5*, A1. https://doi.org/10.1051/swsc/2015001

Burke, W. J. (2018). Thermospheric dynamics during the March 1989 magnetic storm. Balkan, Black sea and Caspian sea regional network for space weather studies. https://doi.org/10.31401/SunGeo.2018.02.07

Codrescu, S. M., Rowland, W. F., Plummer, T. M., Vanier, B. A., Berger, T. E., & Codrescu, M. V. (2020). Feasibility of near-real-time GOLD data products. *Journal of Geophysical Research: Space Physics*, *125*(6), e2020JA027819. https://doi.org/10.1029/2020JA027819

Crowley, G., Knipp, D. J., Drake, K. A., Lei, J., Sutton, E., & Lühr, H. (2010). Thermospheric density enhancements in the dayside cusp region during strong B_Y conditions: Thermospheric density enhancements. *Geophysical Research Letters*, *37*(7), L07110. https://doi.org/10.1029/2009GL042143

Dang, T., Li, X., Luo, B., Li, R., Zhang, B., Pham, K., et al. (2022). Unveiling the space weather during the Starlink satellites destruction event on 4 February 2022. *Space Weather*, *20*(8), e2022SW003152. https://doi.org/10.1029/2022SW003152

Dominique, M., Hochedez, J.-F., Schmutz, W., Dammasch, I. E., Shapiro, A. I., Kretzschmar, M., et al. (2013). The LYRA instrument onboard PROBA2: Description and in-flight performance. *Solar Physics*, *286*(1), 21–42. https://doi.org/10.1007/s11207-013-0252-5

Eastes, R. W., McClintock, W. E., Burns, A. G., Anderson, D. N., Andersson, L., Codrescu, M., et al. (2017). The global-scale observations of the limb and disk (GOLD) mission. *Space Science Reviews*, *212*(1–2), 383–408. https://doi.org/10.1007/s11214-017-0392-2

Emmert, J. T., Drob, D. P., Picone, J. M., Siskind, D. E., Jones, M., Mlynczak, M. G., et al. (2020). NRLMSIS 2.0: A whole atmosphere empirical model of temperature and neutral species densities. *Earth and Space Science*, *7*(3), e2020EA001321. https://doi.org/10.1029/2020EA001321

Emmert, J. T., Warren, H. P., Segerman, A. M., Byers, J. M., & Picone, J. M. (2017). Propagation of atmospheric density errors to satellite orbits. *Advances in Space Research*, *59*(1), 147–165. https://doi.org/10.1016/j.asr.2016.07.036

ESA. (2022). ESA's annual space environment report. GEN-DB-LOG-00288-OPS-SD Retrieved from https://www.sdo.esoc.esa.int/environment_report/Space_Environment_Report_latest.pdf

Fang, T.-W., Kubaryk, A., Goldstein, D., Li, Z., Fuller-Rowell, T., Millward, G., et al. (2022). Space weather environment during the SpaceX Starlink satellite loss in February 2022. *Space Weather*, *20*(11), e2022SW003193. https://doi.org/10.1029/2022SW003193

Forbes, J. M., Marcos, F. A., & Champion, K. S. W. (1978). Lower thermosphere response to geomagnetic activity. *Space Research*, *XVIII*, 173–176. Cited in Berger & Barlier (1981).

Gombosi, T. I., Chen, Y., Glocer, A., Huang, Z., Jia, X., Liemohn, M. W., et al. (2021). What sustained multi-disciplinary research can achieve: The space weather modeling framework. *Journal of Space Weather and Space Climate*, *11*, 42. https://doi.org/10.1051/swsc/2021020

Gopalswamy, N. (2008). Solar connections of geoeffective magnetic structures. *Journal of Atmospheric and Solar-Terrestrial Physics*, *70*(17), 2078–2100. https://doi.org/10.1016/j.jastp.2008.06.010

Hapgood, M., Liu, H., & Lugaz, N. (2022). SpaceX—Sailing close to the space weather? *Space Weather*, *20*(3), e2022SW003074. https://doi.org/10.1029/2022SW003074

Kataoka, R., Shiota, D., Fujiwara, H., Jin, H., Tao, C., Shinagawa, H., & Miyoshi, Y. (2022). Unexpected space weather causing the reentry of 38 Starlink satellites in February 2022. *Journal of Space Weather and Space Climate*, *12*, 41. https://doi.org/10.1051/swsc/2022034

Kornfeld, R. P., Arnold, B. W., Gross, M. A., Dahya, N. T., Klipstein, W. M., Gath, P. F., & Bettadpur, S. (2019). GRACE-FO: The gravity recovery and climate experiment follow-on mission. *Journal of Spacecraft and Rockets*, *56*(3), 931–951. https://doi.org/10.2514/1.A34326

Laskar, F. I., Pedatella, N. M., Codrescu, M. V., Eastes, R. W., Evans, J. S., Burns, A. G., & McClintock, W. (2021). Impact of GOLD retrieved thermospheric temperatures on a whole atmosphere data assimilation model. *Journal of Geophysical Research: Space Physics*, *126*(1), e2020JA028646. https://doi.org/10.1029/2020JA028646

Laskar, F. I., Pedatella, N. M., Codrescu, M. V., Eastes, R. W., & McClintock, W. E. (2022). Improving the thermosphere ionosphere in a whole atmosphere model by assimilating GOLD Disk temperatures. *Journal of Geophysical Research: Space Physics*, *127*(1), e2021JA030045. https://doi.org/10.1029/2021JA030045

Licata, R. J., Tobiska, W. K., & Mehta, P. M. (2020). Benchmarking forecasting models for space weather drivers. *Space Weather*, *18*(10), e2020SW002496. https://doi.org/10.1029/2020SW002496

Lin, D., Wang, W., Garcia-Sage, K., Yue, J., Merkin, V., McInerney, J. M., et al. (2022). Thermospheric neutral density variation during the "SpaceX" storm: Implications from physics-based whole geospace modeling. *Space Weather*, *20*(12), e2022SW003254. https://doi.org/10.1029/2022SW003254

Liu, J., Burns, A. G., Wang, W., & Zhang, Y. (2020). Modeled IMF B_y effects on the polar ionosphere and thermosphere coupling. *Journal of Geophysical Research: Space Physics*, *125*(3), e2019JA026949. https://doi.org/10.1029/2019JA026949

Lu, G. (2017). Large scale high-latitude ionospheric electrodynamic fields and currents. *Space Science Reviews*, *206*(1–4), 431–450. https://doi.org/10.1007/s11214-016-0269-9

Lucas, G. (2021). *Pymsis (v0.4.0)*. Zenodo. https://doi.org/10.5281/zenodo.6299691

Oliveira, D. M., Zesta, E., Mehta, P. M., Licata, R. J., Pilinski, M. D., Tobiska, W. K., & Hayakawa, H. (2021). The current state and future directions of modeling thermosphere density enhancements during extreme magnetic storms. *Frontiers in Astronomy and Space Sciences*, *8*, 764144. https://doi.org/10.3389/fspas.2021.764144

Palmerio, E., Kilpua, E. K. J., Mostl, C., Bothmer, V., James, A. W., Green, L. M., et al. (2018). Coronal magnetic structure of earthbound CMEs and in situ comparison. *Space Weather*, *16*(5), 442–460. https://doi.org/10.1002/2017SW001767

Pham, K. H., Zhang, B., Sorathia, K., Dang, T., Wang, W., Merkin, V., et al. (2022). Thermospheric density perturbations produced by traveling atmospheric disturbances during August 2005 storm. *Journal of Geophysical Research: Space Physics*, *127*(2), e2021JA030071. https://doi.org/10.1029/2021JA030071

Picone, J. M., Hedin, A. E., Drob, D. P., & Aikin, A. C. (2002). NRLMSISE-00 empirical model of the atmosphere: Statistical comparisons and scientific issues. *Journal of Geophysical Research*, *107*(A12), SIA15-1–SIA15-16. https://doi.org/10.1029/2002JA009430

Pilinski, M., Crowley, G., Sutton, E., & Codrescu, M. (2016). Improved orbit determination and forecasts with an assimilative tool for satellite drag specification. In *AMOS conference poster*. Retrieved from https://amostech.com/TechnicalPapers/2016/Poster/Pilinski.pdf

Pulkkinen, T., Gombosi, T. I., Ridley, A. J., Toth, G., & Zou, S. (2021). The space weather modeling framework goes open access. *Eos*, *102*. https://doi.org/10.1029/2021EO158300

Robbrecht, E., Berghmans, D., & Van der Linden, R. A. M. (2009). Automated LASCO CME catalog for solar cycle 23: Are CMEs scale invariant? *The Astrophysical Journal*, *691*(2), 1222–1234. https://doi.org/10.1088/0004-637X/691/2/1222

Rosen, D. R., & Johnson, D. L., & the NOAA Service Assessment Team. (2004). *Service assessment: Intense space weather storms October 19—November 07, 2003*. U.S. Department of Commerce, National Oceanic and Atmospheric Administration.

Storz, M. F., Bowman, B. R., Branson, M. J. I., Casali, S. J., & Tobiska, W. K. (2005). High accuracy satellite drag model (HASDM). *Advances in Space Research*, *36*(12), 2497–2505. https://doi.org/10.1016/j.asr.2004.02.020

Sutton, E. K., Forbes, J. M., Nerem, R. S., & Woods, T. N. (2006). Neutral density response to the solar flares of October and November, 2003. *Geophysical Research Letters*, *33*(22), L22101. https://doi.org/10.1029/2006GL027737

Thiemann, E. M. B., & Dominique, M. (2021). PROBA2 LYRA occultations: Thermospheric temperature and composition, sensitivity to EUV forcing, and comparisons with Mars. *Journal of Geophysical Research: Space Physics*, *126*(7), e2021JA029262. https://doi.org/10.1029/2021JA029262

Thiemann, E. M. B., Dominique, M., Pilinski, M. D., & Eparvier, F. G. (2017). Vertical thermospheric density profiles from EUV solar occultations made by PROBA2 LYRA for solar cycle 24: LYRA thermospheric profiles. *Space Weather*, *15*(12), 1649–1660. https://doi.org/10.1002/2017SW001719

10.1029/2022SW003330

Tobiska, W. K., Bowman, B. R., Bouwer, S. D., Cruz, A., Wahl, K., Pilinski, M. D., et al. (2021). The SET HASDM density database. *Space Weather*, *19*(4), e2020SW002682. https://doi.org/10.1029/2020SW002682

Zhang, S., He, L., & Wu, L. (2020). Statistical study of loss of GPS signals caused by severe and great. *Geomagnetic Storms*, *125*(9), e2019JA027749. https://doi.org/10.1029/2019JA027749

Zhang, Y., Paxton, L. J., Schaefer, R., & Swartz, W. H. (2022). Thermospheric conditions associated with the loss of 40 Starlink satellites. *Space Weather*, *20*(10), e2022SW003168. https://doi.org/10.1029/2022SW003168